

Matrix lives?

Controversial futurist Ray Kurzweil is well-known for making bold statements about the near future. He correctly predicted the rise of the Internet; that a computer would beat a grandmaster in chess by 1998; that LCD displays would soon surpass CRT displays. But his latest Web site prediction takes some serious swallowing.

Kurzweil argues that the future will resemble *The Matrix*. The exponentially increasing computer technology, plus an increasing ability to comprehend the mechanics of the human brain (to then reverse-engineer it), will allow us to have "sentient...programs—the



ability to directly download capabilities into the human brain, and the creation of virtual realities indistinguishable from the real world."

He points out that the raw computing power of the brain (estimated to be 20 quadrillion instructions per second) will be soon be available and inexpensive.

Moore's Law, he opines, will continue indefinitely, as new computing paradigms replace silicon chips. Virtual Reality will then be pervasive, and human beings will spend most of their time living and interacting in simulated environments. "We will make a hundred years of progress, in 25 years. The next 10 years will be like 20, and the following ten years will be like 40. The 21st century will therefore be like 20,000 years of progress."

E.T. might call home

SETI@Home is a popular distributed network, set up to look for signs of extra-terrestrial intelligence. The network has picked up millions of radio signals—indicators of intelligence—since 1999.

Over four million people have downloaded the screen-saver onto their personal computer. When idle the screen-saver is activated, and crunches the data that it receives from SETI. Each packet is a 100-second segment of noise, collected by the Arecibo radio telescope. SETI is notified whenever a signal appears to have been emitted from a single source.



Over 350 million such signals have been spotted in the radio data. But in the 24 hours available, the astronomers are able to revisit a maximum of 150 signals. According to Dan Werthimer, the chief scientist for the SETI@home project, there is a one in 10,000 chance that an extraterrestrial intelligence was the source of a candidate signal.

COPA termed 'anti-adult': ACLU

A three-judge panel of the Third U.S Circuit Court of Appeals, has ruled that the Child Online Protection Act (COPA) restricted free speech, by barring Web site operators from posting information inappropriate for minors, unless they limited the site to adults. The ruling upheld an injunction blocking the government from enforcing the law.

The court stated that, in practice, the law made it too

difficult for adults to view material protected by the First Amendment, including many non-pornographic sites. The law has never been enforced. The American Civil Liberties Union (ACLU), which initiated the legal challenge, said the law would have made it a crime to communicate a whole range of information to adults. The government may ask the Third Circuit to rehear the case or appeal to the U.S Supreme Court.

quoteworthy

"In almost all cases, virus writers were computer-obsessed males between the ages of 14 to 34 years. They have a chronic lack of girlfriends, are usually socially inadequate and are drawn compulsively to write self-replicating codes. It's a form of digital graffiti to them."

Jan Hruska, Chief Executive of British-based Sophos, the world's fourth-largest anti-virus solutions provider, in an interview to Reuters

"Just like cellphones and cellular infrastructure unleashed telecoms, we believe in this decade wireless notebooks are going to unleash computing"

Intel Chief Operating Officer Paul Otellini, at an event in Tokyo

tomorrow's technology

Fuel cells to power mobiles

To power notebooks and other mobile devices, technology companies are looking to fuel cells. NEC and Hitachi are trying to device such cells for the mobile segment. The rechargeable batteries of today, do not offer the power or the capacity that will satiate the hunger of the mobile devices

of tomorrow. Both companies demonstrated their offerings at the Nano Tech 2003 show, recently held in Japan. Toshiba also announced to the world a successful development of a prototype fuel cell. They plan to commercialise their offerings by 2005.

NEC's system uses 'car-

bon nano horns' as an electrode, that can pump 12W of power. It is trying to incorporate monitoring and management systems to ensure smooth and stable power generation. Hitachi is looking at the PDA market, and employs metallic nano particles as electrode catalysts.